U208: a finite-history decoder fold for the consistency of Peano arithmetic

Comparing the five U209 submissions and saving another state

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Abstract

The five complete 209-state submissions in the requested ProveIt snapshot represent two rooted transition-graph isomorphism classes, rather than five different machine designs. All execute the same 3,909 primitive instructions and the same 5,083 ordinal control edges. We compare their exact tables, source arrangements, reconstruction methods, and proof boundaries.

The new construction deletes native state 211 by sharing native state 172. Two entry invariants, established from complete predecessor histories of length at most three, distinguish the required behaviors. The resulting machine has **208 working states and 416 transitions**, including blank initialization. Exactly three retained transition rows change, two rows are deleted, and no source instruction or source address changes.

Four complete local cases prove halting equivalence with the supplied U209 parent on every common initial tape from state 0. Old steps expand to one, three, or seven new steps. A separate fresh symbolic derivation verifies all 5,084 arbitrary-value primitive contracts and all 416 literal rows. The exact 48,140-step blank initialization is preserved. The same reduction applies to all five proposals; a second, structurally distinct 208-state variant is also supplied.

The new local preservation theorem is independent of the logical calibration. The intended Con(U) and Con(PA) endpoint composes it with the retained conventional proof-search and interpretation arguments. This packet does not provide a new end-to-end Lean, Rocq, MM0, or exported PA proof certificate. It also preserves a concrete counterexample to a tempting source-reassociation shortcut, so that the unsuccessful attempt is not mistaken for a second valid reduction.

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1 Result, conventions, and proof status

The primary artifact is `machine/u208.tm`. Its exact conventions are:

Working states	$0, 1, \ldots, 207$
Separate halt label	208
Initial state and head	0 and tape coordinate 0
Alphabet and blank symbol	$\{0, 1\}$, with blank 0
Tape	One two-way infinite tape, initially all zero
Transition rows	416; every working state has both binary rows
Moves	Left or right on every transition
Initialization	Included in the 208 working states

A row $q\ b\ w\ D\ r$ means: in state q, read b, write w, move in direction D, and enter r. Entering the halt label terminates after the write and move. The header `#! start 0` is not a transition.

Native labels retain the historical names, with native halt 254. In particular, *delete native state 211* is a label-level instruction, not a state-count claim. The exact bijection in `machine/state_map.json` converts native labels to the dense table. Native 172 becomes dense 165. All 208 working states are reachable in the transition graph from state 0. This graph property does not assert that every state is visited on the blank run.

Let M be the exact Plain U209 parent retained in `inputs/u209.native.tm`, and let N be the new U208. The principal new theorem is

$$\mathrm{Halt}(M, T, 0) \iff \mathrm{Halt}(N, T, 0)$$

for every common binary tape T, with a common initial head position. The last argument indicates the initial state. Section 4 proves this statement directly, using only the literal tables.

The arithmetic instance needed for the research goal is

$$\mathsf{PA} \vdash \mathrm{Halt}(N, \mathrm{blank}) \iff \mathrm{Halt}(M, \mathrm{blank}). \tag{1}$$

The finite-history and bounded-block argument supplies an ordinary PA-formalizable proof of this equivalence. Combining it with the retained U209 calibration yields the intended endpoint

$$\mathsf{PA} \vdash \neg\,\mathrm{Halt}(N, \mathrm{blank}) \iff \mathrm{Con}(\mathsf{U}) \iff \mathrm{Con}(\mathsf{PA}). \tag{2}$$

The inherited dependencies and the evidence for each layer are identified in Section 7. Equation (2) is not presented as a newly checked proof-assistant declaration.

Claim	Evidence in this packet
Exact 208-state object	Literal native/dense tables, totality, graph census, checked relabeling, deterministic reconstruction, hashes.
New U209-to-U208 theorem	Complete incoming inventories, independent suffix-history enumeration, four whole-tape macros, 414 ordinary row identities, and the run-induction proof.
Canonical primitive runtime	Fresh unary/list-inductive derivation of all 5,084 contracts from the exact new table; 416/416 row coverage.
Blank entry	Exact full-tape and full-trace comparison through the first canonical frame satisfying the proved header bound.
Consistency calibration	Retained mathematical proof-search, syntax coding, protected-startup, and U/PA finite-refutation translations.
Formal certificate	No new end-to-end PA, Lean, Rocq, or MM0 certificate.

Neither consistency nor unconditional nonhalting is assumed in the local equivalence. Actual nonhalting follows from Con(PA) once the retained calibration is accepted. No global minimality or historical priority claim is made.

2 Comparison of the supplied U209 proposals

2.1 Pinned inventory

The repository snapshot used here is [1]

`343da82d7384d10ec9fd1c17361fc923b2d1d117`.

All downloaded archive bytes were checked against their Git blob identities. Full SHA-256 values, source URLs, archive sizes, and exact member identities are recorded in `reference/source_manifest.json`, `reference/archive_hashes.json`, and `comparison/provenance.json`. The five complete U209 archives are denoted by the short names in Table 1. A sixth U209-named ZIP, `U209_ConPA_Joint_2026-10-29 (1).zip`, contains only a PDF.

Submission	Graph class	Depth	Zero run	Dense SHA-256 prefix
Plain	A	42	6	`65ccb606c8e970ad`
Compatibility	B	41	6	`bda91bd40880f96e`
Compatible Folds	A	42	6	`5ded628d692ff442`
Composed	A	42	6	`3a69998d818b6e63`
Joint	A	42	6	`5ded628d692ff442`

Table 1: Five complete submissions, two rooted action-labeled graph classes. Every table has 209 graph-reachable working states and 418 rows.

The Plain archive is `U209_ConPA_2026-10-29.zip`; the other names insert the displayed qualifier after `ConPA`. Compatible Folds and Joint have byte-identical native and dense tables. Plain and Composed are nontrivial state renamings of that same machine. Compatibility is not isomorphic to them as a rooted action-labeled transition graph.

The comparison canonically numbers working states by breadth-first discovery from state 0, processing the read-zero edge before the read-one edge. Each canonical row retains the read symbol, written symbol, move, and target; halt receives a separate distinguished symbol. Because all working states are graph-reachable and binary transition labels are fixed, matching canonical encodings give an exact rooted isomorphism. For each matching pair, the program additionally reconstructs the state bijection and checks every transition under it. Thus the classification is stronger than a comparison of file hashes.

The four class-A sources are byte-identical, with SHA-256

`cddbed7f20e7a8b085b7ca0e76811d07b021522375643b5d1134eec0d2d23164`.

Compatibility's source has SHA-256

`912724f29b0de28b3326c710fdf1f3863ebd07058c6db1a9b4bbaffb3ae5e3f3`.

Nevertheless, all five have the same 3,909 ordered primitive opcodes and operands, the same 5,083 ordinal successor edges, and 197 distinct syntactic subtrees. The different depth and addressing describe different layouts of the same program, not a different proof calculus.

The PDF-only supplement is a different report from the full Joint report. Its printed native/dense machine hashes identify the Plain representative. It therefore does not add a sixth executable design or a third graph class. The original supplement is retained for audit.

2.2 How the constructions differ

All five submissions combine earlier savings associated with native states 180, 181, and 218. Those earlier U211 savings could not simply be overlaid: the shared helper states required different incoming tape conditions. The U209 packets solve this compatibility problem with different descriptions of source reassociation and helper allocation.

Plain and Composed expose a sequence of 30 closed source rotations. Composed additionally presents a label permutation and is particularly explicit about the complete logical ancestry retained from U222. Joint organizes the main change as one reassociation of a 3,096-primitive region using 18 internal selectors, followed by three smaller rotations. Compatible Folds gives the same final source and literal machine as Joint. These are useful differences in proof organization, but they are not additional state-count improvements.

Compatibility yields the distinct class B. Its construction starts from Research U211, passes through U210, and permits a shared backward helper to retain its read-zero behavior while changing its read-one source-decoder behavior. Its final 180 fold uses native 216 as the representative, where class A uses native 163. This creates a different decoder association with maximum depth 41.

For subsequent work, class A is the larger consensus among the supplied packets; Plain is a convenient exact parent for the three-row patch below. Joint has a compact grouped builder, and Composed preserves a particularly clear account of the recovered logical calibration. Compatibility deserves separate analysis because it is a different machine, even though its primitive program is identical. Neither the smaller depth nor the number of symbolic inference steps establishes a faster complete proof search.

2.3 Fresh replay and the meaning of the evidence

All five supplied quick replays passed in isolated working copies. The complete symbolic runtime replay was also rerun for one representative of each graph class:

Representative	Primitive contracts	Rows covered	Symbolic inferences
Joint, class A	5,084	418/418	1,250,049
Compatibility, class B	5,084	418/418	1,248,748

Both fresh per-contract records matched the supplied records. The logs and receipts are retained under `evidence/`. These counts describe the work performed by the verification engine, not the duration of the eventual consistency search.

All submissions distinguish executable inductive evidence from a new end-to-end formal certificate. The comparison retains that distinction. In particular, the formal K215/ZFC theorem in the supplied external repositories concerns a different machine and a different calibration.

3 The new construction: three changed rows

3.1 Literal recipe

Define a state map on native labels by

$$\rho(211) = 172, \qquad \rho(q) = q \quad (q \neq 211).$$

Starting from the exact Plain U209 table, redirect every target through ρ, remove both rows of state 211, and set

$$N(172, 0) = 0\,L\,214. \tag{3}$$

No other action is altered. Table 2 is the complete difference.

State	Read	Old action	New action	Reason
163	0	$0\,L\,211$	$0\,L\,172$	Redirect target
163	1	$1\,R\,211$	$1\,R\,172$	Redirect target
172	0	$0\,L\,67$	$0\,L\,214$	New dispatch
211	0	$0\,R\,157$	Removed	Merged state
211	1	$1\,R\,184$	Removed	Merged state

Table 2: Complete native-table patch. Exactly three retained rows change.

The retained row 172/1 is $1\,R\,184$, equal to the deleted row 211/1. The difficulty is the two read-zero behaviors: one must move left to 67, the other right to 157. Reading a short portion of the already written tape distinguishes them without another state.

After deletion, sort the 208 remaining native working labels increasingly and rename them $0, \ldots, 207$; rename native halt 254 to 208. The builder checks both directions of the resulting table correspondence. The exact reconstruction uses no search, source optimizer, solver, or external dependency.

3.2 Why this differs from the previous reductions

The new optimization does not reassociate the source, reassign a source subtree, or modify the numerical proof calculus. Its required information comes from the histories of entrances to existing states. In particular, state 211 is entered both from the left and from the right; examining only the direction and bit written by the immediate predecessor loses a useful conditional invariant.

The discovery search enumerated finite predecessor histories, converted them to disjunctions of known tape cells, and screened possible shared-state dispatches. The delivered proof does not depend on the discovery search. It reduces the successful result to the five explicit incoming inventories and four explicit local cases below. Novelty is claimed relative to the audited incoming packets, not as an independent historical priority claim.

4 Complete preservation proof

4.1 Two entry invariants of the old machine

For a configuration with head at h, let $t(j)$ denote the tape bit at offset j, that is, at coordinate $h + j$. All inventories in this subsection refer to the *old* machine M. The full incoming-edge lists are:

Target	All incoming transitions
172	$116/0 \mapsto 0\,R\,172$
116	$216/1 \mapsto 1\,R\,116$
211	$163/0 \mapsto 0\,L\,211$, $163/1 \mapsto 1\,R\,211$
163	$132/1 \mapsto 1\,R\,163$, $199/1 \mapsto 1\,R\,163$, $214/0 \mapsto 0\,R\,163$
214	$204/1 \mapsto 1\,R\,214$

None of these states is the initial state 0. Consequently, every visit on a run starting at 0 has the necessary predecessor history, regardless of the initial tape.

Lemma 4.1 (Guard at old 172)**.** *Every reachable old configuration in state 172 has*

$$t(-2) = 1, \qquad t(-1) = 0.$$

Proof. The only entrance to 172 is 116/0, which writes zero immediately left of the new head. The only entrance to 116 is 216/1, which writes one one cell farther left. Both moves are rightward and neither step changes the other cell. This establishes the two asserted bits. □

Lemma 4.2 (Conditional guard at old 211)**.** *Every reachable old configuration in state 211 scanning zero has* $t(-1) = 1$.

Proof. If the last step was 163/1, it wrote one and moved right, so the conclusion is immediate.

Otherwise the last step was 163/0, which wrote zero and moved left. The scanned cell at old 211 is then the cell immediately left of the previous head at 163. An entrance to 163 through 132/1 or 199/1 would have written one in this cell, contradicting the present read-zero condition. The entrance to 163 must therefore have been 214/0, a right move after writing zero. The only entrance to 214 is 204/1, which wrote one in the cell now immediately left of the head at 211. The intermediate two steps do not change that cell. Hence $t(-1) = 1$. □

The argument for Lemma 4.2 uses at most three predecessor transitions. The independent checker enumerates all such suffix histories and all assignments to their touched cells; it does not sample a finite prefix of the blank run. Unread exterior cells remain arbitrary.

4.2 The four local cases

Besides the changed row, the relevant new rows are

$$\begin{array}{llll} 214/0 & \mapsto & 0\,R\,163, & 214/1 \mapsto 1\,R\,216, \\ 163/0 & \mapsto & 0\,L\,172, & 216/0 \mapsto 0\,R\,157, \\ 157/0 & \mapsto & 0\,L\,67, & 172/1 \mapsto 1\,R\,184. \end{array}$$

In the following traces, $q@j$ means state q with head offset j relative to its position at the start of the old step.

Old 211, scanning zero. The known window is $t(-1) = 1, t(0) = 0$. Starting the new machine in the representative 172 gives

$$172@0 \longrightarrow 214@{-1} \longrightarrow 216@0 \longrightarrow 157@1.$$

The three steps read $0, 1, 0$, respectively. Each writes the bit it reads. The old single step $211/0 \mapsto 0\,R\,157$ has exactly this final state, head position, and whole tape.

Old 172, scanning zero. The known window is $t(-2) = 1, t(-1) = 0, t(0) = 0$. The new seven-step path is

$$\begin{aligned} 172@0 &\longrightarrow 214@-1 \longrightarrow 163@0 \longrightarrow 172@-1 \\ &\longrightarrow 214@-2 \longrightarrow 216@-1 \longrightarrow 157@0 \longrightarrow 67@-1. \end{aligned}$$

Its reads are $0, 0, 0, 0, 1, 0, 0$. Again, every step writes the bit it reads. The endpoint is precisely that of the old step $172/0 \mapsto 0\,L\,67$.

This path temporarily re-enters new state 172. It is not an unproved recursive call: the complete path has been expanded, and the second visit sees the other guard, so it terminates after the stated seven steps.

Old 172 or 211, scanning one. Both old actions are $1\,R\,184$. The new representative has the same action. Both cases take one new step, require no additional neighboring cell assumption, and preserve the tape.

Thus the exhaustive case table is

Old state	Read	Required old window	New steps	Final state/head
172	0	$t(-2)t(-1)t(0) = 100$	7	67@−1
211	0	$t(-1)t(0) = 10$	3	157@1
172	1	$t(0) = 1$	1	184@1
211	1	$t(0) = 1$	1	184@1

No special case reads an unspecified exterior cell or encounters halt internally. The checker compares the entire represented tape after each macro, not merely the cells thought to matter.

4.3 Global simulation and its converse

Theorem 4.3 (All-common-tapes preservation)**.** *For every binary tape and common initial head position, the runs of M and N from state 0 halt together. At aligned macro boundaries, their tapes and heads are equal and their states are related by ρ.*

Proof. The initial configurations satisfy the relation because $\rho(0) = 0$. Suppose the relation holds at an old-step boundary. If the old state is neither 172 nor 211, the corresponding new transition is exactly its quotient under ρ. All 414 such old rows are checked literally. If the old state is 172 or 211, the relevant guard follows from Lemmas 4.1–4.2, and one of the four proved macros restores the relation. Every macro takes a positive number of steps, at most seven, and none halts internally.

Induction therefore constructs an aligned new execution for every finite old execution. If the old execution halts, its aligned new execution halts. If the old execution is infinite, positive macro lengths produce infinitely many new steps, none with an internal halt. Determinism makes this constructed execution the unique new execution. The new machine therefore cannot halt when the old machine does not. This proves both implications. □

The guards concern old boundary configurations. They need not hold at every intermediate new configuration, and the argument does not assume that they do. This distinction is essential because the seven-step case passes through the representative state at an intermediate head position.

Corollary 4.4 (Exact time accounting)**.** *For any aligned finite prefix, let $n_{172,0}$ and $n_{211,0}$ count old steps using the two zero rows. Then*

$$T_N = T_M + 6n_{172,0} + 2n_{211,0}, \qquad T_M \le T_N \le 7T_M.$$

The same identity holds for a halting run.

This comparison is directly with the supplied Plain U209 table. Unlike the earlier source-reassociation constructions, the new reduction changes no addresses, so no separate source-layout stage lies outside the time bound. It is a reduction of finite-control size, not a claimed speedup.

4.4 Arithmetization

For the blank-input theorem, configurations and finite runs have ordinary number codes. The two entry guards are finite predecessor facts; the four macros are literal constant-size computations. PA can prove the simulation by induction on the coded old run length, defining the aligned new length $\tau(n)$ with

$$n \le \tau(n) \le 7n.$$

The converse can also be expressed entirely with finite prefixes: if the new machine halted after s steps but the old machine had not halted in its first s steps, the constructed old-prefix simulation would extend the new run for at least s steps without a halt, a contradiction. No appeal to actual consistency or to a soundness principle is required.

The external theorem allows arbitrary binary tapes as functions on the integers. The internal PA statement quantifies coded finite input data and finite computation histories, including the blank instance. It does not introduce second-order quantification over arbitrary tape functions into PA. The packet supplies the finite certificates and the mathematical induction, not an exported derivation in a PA proof language.

5 Portability to both structural families

The same mechanism is present in every supplied U209 table. Three native role labels vary:

Submission	Entry E	Selector H	Follower F
Plain	163	214	216
Compatibility	216	214	163
Compatible Folds	163	217	215
Composed	163	216	215
Joint	163	217	215

Here E is the state whose two rows enter old 211; $H/0$ enters E, $H/1$ enters F, and $F/0$ enters 157. The checked incoming inventories establish the same guards. In every case delete 211, redirect its targets to 172, and replace 172/0 by $0\,L\,H$.

The role-based checker derives these roles from each table, verifies the complete incoming inventories, and replays the same four macros. All five checks pass. The construction therefore does not rely on choosing a particularly favorable state naming among the equivalent submissions.

`machine/compatibility_variant/` contains the class-B 208-state table, source, and relabeling. Its source depth remains 41; the primary candidate's depth remains 42. The primary candidate received

the complete fresh final-table symbolic runtime replay described next. The alternative is certified by its local all-tape reduction and the freshly replayed Compatibility U209 parent; we do not mislabel that as a separate full symbolic replay of the alternative.

6 Verification against the exact new table

6.1 Independent local certificates

Two separately written checkers verify the new fold. `checks/check_local.py` rebuilds the precise table and checks the specified incoming inventories and all four macro endpoints. `checks/check_local_independent.py` independently enumerates complete two-step histories for state 172 and three-step histories for state 211 scanning zero. For each history it enumerates every assignment on touched cells and leaves the exterior opaque. This verifies the guards without importing the discovery search or the primary checker's invariant logic.

Both checkers validate the dense/native bijection, state census, and ordinary rows. Both pass with normal Python and optimized Python. The first rejects eleven deliberately damaged constructions, including corrupted predecessor tags, a wrong helper, a wrong move, a wrong shared one branch, and a changed ordinary transition.

6.2 Unchanged program and exact blank startup

The complete source file is byte-identical to the Plain parent. The checks confirm:

Source quantity	Value
Ordered primitive occurrences	3,909
Decrement occurrences with two guarded branches	1,175
Primitive contracts	5,084
Nonhalt ordinal control edges	5,083
Distinct syntactic subtrees	197
Maximum address length	42
Maximum consecutive zero run	6
Changed source addresses	0

Two source-control resolvers, using interval ancestry and binary increment/cut respectively, agree. No source-level semantic claim is needed for the direct local theorem, but the unchanged source is useful when connecting the machine to its inherited proof search.

The old and new blank computations agree in every state, head position, transition, and tape cell for the complete first 48,140 steps. They then reach native state 84 with head -2, 120 one-cells, source ordinal 136, and header distance 56. The nine logical registers are

$$(1, 0, 13, 0, 0, 1, 0, 0, 1),$$

followed by 43 spare blocks, each of value 1. This is the first complete canonical frame with header distance at least 56; it is *not* the first visit to the dispatcher. The dispatcher is visited earlier during initialization. None of the changed rows occurs in this verified prefix.

The startup checker records the exact one-cell set and a full trace digest. It also checks the finite source bridge from the inherited protected seed to that canonical frame. The general justification of the protected tail and the fairness of the proof enumeration remain part of the retained calibration.

6.3 Fresh arbitrary-value primitive derivation

The final 208-state literal table was passed through the retained whole-tape symbolic-induction engine, whose two inference modules are pinned byte-for-byte. Its generic primitive configuration contains:

- nine arbitrary natural primary registers;
- an arbitrary natural extra header gap;
- an arbitrary nonempty finite list of spare register blocks;
- both zero exteriors and the complete program-counter/header region.

The register block of value v is $B(v) = 0\,1^{v+1}$. The engine proves unary traversals by induction on arbitrary run lengths and proves four generic block/list traversals from the literal table. Its list normalization uses the finite-word identity $C(A)1 = 1B(A)$, where $C(v) = 10\,1^v$; the identity follows by induction on the finite list A.

Every primitive contract is then rederived from the new table. Nonhalting primitive endpoints compare the complete symbolic tapes, not just register values. Duration expressions have thirteen affine slots: a constant, the gap, nine primary values, the first spare value, and the remaining spare-list width. Every coefficient is nonnegative and every constant term is positive. The resulting census is

Fresh U208 result	Value
Primitive contracts	5,084
Literal rows covered	416/416
Symbolic inference applications	1,251,865
Minimum affine-duration constant	5
Generic block/list laws	4

These are executable mathematical induction checks, not a bounded register sample and not a proof-assistant kernel. The local theorem of Section 4 is an additional, simpler route to preservation and does not depend on the engine's implementation.

6.4 Independent finite execution

An independent C++ executor checked 15,252 concrete primitive cases, covering all source sites and both decrement branches under three register and spare assignments. It performed 6,778,830 literal transitions in those cases. All complete endpoints matched the source semantics. A separate two-million-step blank regression did not halt. The finite tests are implementation checks; the absence of a halt in that prefix is not evidence of a proved consistency statement.

The finite executor is optional for reproduction because the standard archive replay needs only Python. Its source and receipt are supplied. No compiler is needed to check the finite-history proof.

7 The inherited logical endpoint

7.1 The specified unary theory

The supplied unary-power-cover report [2] defines U in equality-free classical first-order logic over a nonempty domain, with one binary relation written $\in$. Define

$$x \subseteq a \quad :\Longleftrightarrow \quad \forall t\,(t \in x \longrightarrow t \in a).$$

For every formula $\varphi(x)$ with no free variable other than x, the single axiom schema is

$$\forall a\,\exists b\,\left[\forall x\,(x \in b \longrightarrow \varphi(x)) \;\wedge\; \forall x\,(x \subseteq a \longrightarrow (\varphi(x) \longrightarrow x \in b))\right].$$

The source variable a is not a free matrix parameter. Arbitrarily many bound variables are allowed; this is not one-variable first-order logic. The assertion is weak covering, rather than an exact power filter for arbitrary sources.

The supplied paper gives conventional mutual interpretations and formula-specific finite-proof translations establishing the PA-provable equiconsistency of U and PA. Its lower interpretation first derives representative induction, before successor totality. It then develops comparison, least failed numerical tuples, bounded numerical comprehension, successor constructions, numerical powers, saturated graph codes, and canonical finite histories. This order addresses the possible circularity in using finite constructions to obtain induction.

Its upper interpretation uses binary bit membership on natural numbers. PA proves the required finite comprehension one matrix at a time. This does not require a uniform truth predicate for all arithmetic formulas. Our independent reading found no concrete circularity or illicit free matrix parameter in the examined arguments. The present work does not implement or verify a general arbitrary-PA-proof translator.

7.2 What the literal machine actually searches

The native numerical calculus is calibrated to unrestricted weak covering W, which allows additional matrix parameters, together with irreflexivity IR. It is not the separate positive dyadic theorem stream described in the unary archive. Unary U embeds into $\mathsf{W} + \mathsf{IR}$, while the bit-membership interpretation in PA handles unrestricted matrices and proves irreflexivity. Combined with the lower interpretation into U, the retained finite-refutation translations connect all three consistency statements.

The inherited numerical presentation uses dual Cantor pairing, low-code truth/falsity aliases, seven logical rows, a dominated propositional row, modus ponens, generalization, and a gated covering/vacuity bundle. The code-zero correction, freshness conditions, exceptional variable-role cases, and proof enumeration are substantive parts of the original calibration. They have not been simplified away by the new state fold. The full Plain parent packet, including its nested dependencies and attribution, is retained unchanged in `reference/`.

Thus the logical chain for the new table is:

1. the exact new fold preserves blank halting of the supplied U209;
2. the supplied U209 program searches the retained numerical refutation calculus with its protected initialization and fair enumeration;
3. that calculus is calibrated to unrestricted covering plus irreflexivity;

4. the retained finite-proof translations identify its consistency with Con(U) and Con(PA) inside PA.

Only the first link is newly proved by the finite-history argument. The fresh primitive runtime and startup checks strengthen the executable connection, but do not replace the other logical links.

7.3 Relation to the K215/ZFC proofs

The supplied m254, K215, and k215-pa projects [3, 4, 5] are useful examples of complete machine-to-consistency verification. The k215-pa repository states a checked Metamath Zero theorem in PA for the literal K215 table and ZFC inconsistency. Its certificate does not apply to this U208 table or establish the U-to-PA calibration. We inspected the current repository descriptions; we did not rebuild those large formal developments. The new result is explicitly delivered as a mathematically justified candidate with executable evidence.

8 A rejected source compression, with a counterexample

An additional source search found two appealing closed reassociations:

$$((98, 12), 150) \longmapsto (98, (12, 150)), \qquad ((166, 12), 150) \longmapsto (166, (12, 150)).$$

In native decoder labels, these act at roots 155 and 52. After updating all five occurrences of the first source subtree and all three occurrences of the second, states 85 and 154 acquire identical rows with children $(12, 150)$. Removing 154 then yields another syntactically complete 208-state table.

The entire 3,909-instruction word and all 5,083 source control edges remain unchanged. The final equal-row merge is sound for the already modified table. Nevertheless, the preceding reassociation is invalid as a source-to-runtime transformation, because these physical decoder states also serve the arithmetic engine. The original zero-decrement path includes

$$152/0 \longrightarrow 159/1 \longrightarrow 118/0 \longrightarrow 155/0 \longrightarrow 85/0 \longrightarrow 98/0.$$

The altered association changes which tape cells this runtime path reads. Pure read-preserving right-moving rows and a closed source factor do not by themselves establish that a decoder fragment is used only for decoding.

The failure has an independent finite witness. Start at native state 84, head 0, source ordinal 25 (`CD 1`), with all nine registers zero, gap zero, and spare list [0]. The one-cells are exactly

$$\{-1, 2, 3, 6, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76\}.$$

All other cells are zero. The correct successor is ordinal 28.

Table	Steps to canonical return	Returned instruction
Original U209	494	Ordinal 28, `CD 2`
Rejected source fold	526	Ordinal 44, `CD 4`

The returned frame and registers otherwise agree; the program-counter one is in cell 10 instead of the correct cell 11. The independent standard-library replayer records every literal transition and works with or without Python assertions.

The files under `research/rejected_source_fold/` are expressly rejected exploratory artifacts. This witness refutes their claimed canonical primitive simulation, not a statement about eventual blank

halting. They are not composed with the delivered history-based U208. The failed attempt is useful because it isolates a concrete limitation of source-only compression arguments in this shared-runtime architecture.

9 Reproduction, research limits, and next proof target

9.1 Offline replay

Python 3.10 or later and the standard library suffice. From the extracted archive directory run

```
python3 -B reproduce.py
```

The default command validates the release manifest, rebuilds and checks the literal fold, runs both local checkers normally and under `-O`, compares all five input snapshots, checks the unchanged source and exact blank startup, verifies portability to every parent, and freshly derives all final-table primitive contracts. Fresh cases must equal the saved compressed records. It requires no network, search, optimizer, solver, proof assistant, or C++ compiler.

```
python3 -B reproduce.py --quick
python3 -B reproduce.py --output /tmp/u208-replay
```

Quick mode explicitly skips the fresh universal runtime derivation. It still checks the complete new local preservation certificate; it must not be described as having rerun all arbitrary-value primitive proofs. The default output directory is temporary, leaving the sealed packet unchanged. The full symbolic engine requires enabled assertions; the main driver refuses optimized Python and `PYTHONOPTIMIZE`.

Optional independent finite execution:

```
g++ -std=c++17 -O3 checks/check_finite_runtime.cpp -o /tmp/u208-finite
/tmp/u208-finite machine/u208.native.tm machine/sites.tsv
```

To rebuild this report in a working copy:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

PDF metadata may differ by TeX version or build time. Verify the sealed payload before recompiling it.

9.2 Archive map

Location	Contents
`report.tex`, `report.pdf`	Complete report and compiled PDF.
`paper/`	Full transition table and artifact identity tables.
`machine/`	Primary U208, native/dense bijection, unchanged source, exact transition difference, and the Compatibility variant.
`inputs/`	Frozen primary parent table, source, and runtime label map.
`checks/`	Independent local proofs, source/startup check, five-way comparison, portable fold, and pinned symbolic engine.
`evidence/`	Fresh local, comparison, baseline, runtime, and finite execution records; mathematical audit notes.
`comparison/`	Exact snapshots and member provenance for all five complete U209 submissions.
`reference/`	Unchanged full primary parent archive, unary-theory archive, PDF supplement, repository and archive identities.
`research/`	Optional finite-history discovery and the rejected source-reassociation counterexample.
`licenses/`	Retained licenses and attribution from the source lineage.
`MANIFEST.json`	SHA-256 and size for the sealed payload.

9.3 What remains open

The bounded discovery searches are incomplete. Additional alias and contextual-fold searches on the new U208 did not produce another accepted reduction, and the investigated source shortcut failed. Neither result is a lower bound or a proof that 208 is optimal.

The most valuable verification extension is a small kernel certificate for the exact new local theorem, followed by the complete U-machine calibration. The local theorem is especially suitable for formalization: its premises are five finite incoming inventories, four constant-size macros, and a bounded simulation induction. The more substantial remaining formal work is the inherited numerical-calculus soundness/completeness and the uniform finite-refutation translations for the U/PA interpretation.

For further state reduction, the discovery suggests using disjunctive predecessor histories rather than only immediate edge tags. Any proposed source reassociation should also identify its complete arithmetic-runtime uses before it is accepted. The delivered finite-history fold establishes the requested strict improvement, with its proof and reproduction separated from those future possibilities.

A Exact artifact identities

All hashes in this appendix are SHA-256 of the exact file bytes. Native tables use halt 254; both dense tables use halt 208.

Primary dense U208 `machine/u208.tm`

`926d15c43daebdafd099723f265b073c25bf21a15e4c5593e669ccac6f70f8d0`

Primary native U208 `machine/u208.native.tm`

`36b58fea7a184b1050914170e8f5240c8a13da218dbbfa7e1985cd73d516e5c7`

Exact native U209 parent `inputs/u209.native.tm`

`891455876f2684b2bd2a9118092715590028fbdb227aa7055030536714d8b7a4`

Unchanged primary source `machine/sites.json`

`cddbed7f20e7a8b085b7ca0e76811d07b021522375643b5d1134eec0d2d23164`

Compatibility dense U208 `machine/compatibility_variant/u208.tm`

`7ba8ba5d5a53944e170a8742e34e21a44bf612a1a8de2ce5ddc4e84d8deec675`

Compatibility native U208 `machine/compatibility_variant/u208.native.tm`

`3bc4028dcf445c369623dab733b96819b37a99924b419f02159b2e3954893b1a`

The complete payload manifest records sizes and hashes for all sealed files. The PDF-only supplement and the original primary U209 and unary-theory archives are retained byte-for-byte under `reference/`.

B Complete dense U208 transition table

The initial state is 0, blank is 0, and halt is the separate label 208. Each entry is written symbol, direction, next state. All 416 binary rows are displayed; the halt state has no outgoing rows.

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
0	1 R 92	1 R 0	70	0 L 89	1 R 78	140	0 R 96	1 R 159
1	0 L 1	1 L 2	71	0 R 55	1 L 79	141	0 R 175	1 R 166
2	0 L 3	1 L 2	72	0 R 80	1 R 81	142	0 R 12	1 R 159
3	0 L 5	1 L 2	73	1 R 0	1 R 63	143	0 R 88	1 L 166
4	1 R 6	0 L 4	74	0 R 74	1 R 82	144	0 R 88	1 R 158
5	0 L 9	1 L 2	75	0 L 145	1 L 141	145	0 L 152	1 L 163
6	0 R 6	0 R 10	76	0 R 85	1 R 86	146	0 R 84	1 R 88
7	0 R 190	1 R 159	77	0 R 20	1 R 38	147	0 R 159	1 R 12
8	0 R 14	1 R 13	78	0 R 89	1 L 132	148	0 R 83	1 R 143
9	0 L 15	1 L 2	79	0 L 108	1 R 56	149	0 R 144	1 R 155
10	0 L 16	0 R 10	80	0 R 49	0 R 33	150	0 L 66	1 L 141
11	0 L 26	1 R 128	81	0 R 91	0 L 42	151	0 R 150	1 R 170
12	0 R 19	1 R 20	82	0 R 178	1 R 77	152	0 R 159	1 R 114
13	0 L 89	1 L 31	83	0 R 96	1 R 12	153	0 R 97	1 R 97
14	0 L 11	0 R 22	84	0 R 88	1 R 97	154	0 R 49	0 R 32
15	0 L 23	1 L 2	85	0 R 206	1 R 21	155	0 R 192	1 R 188
16	1 L 1	0 L 1	86	0 R 116	1 R 100	156	0 L 165	1 R 165
17	0 R 24	0 R 32	87	0 R 174	1 R 167	157	0 L 187	1 R 96
18	0 R 98	1 R 113	88	0 R 102	1 R 103	158	0 R 168	1 R 179
19	0 R 28	1 R 29	89	0 R 80	1 R 79	159	0 R 19	1 R 171
20	0 R 30	1 R 31	90	0 R 101	1 R 195	160	0 R 164	1 R 187
21	0 R 24	0 R 59	91	0 R 105	1 R 106	161	0 R 159	1 R 96
22	0 R 33	1 R 22	92	0 R 65	0 R 0	162	0 R 125	1 R 164
23	0 L 34	1 L 2	93	0 R 27	1 R 107	163	0 R 97	1 R 162
24	0 L 35	0 R 0	94	0 R 183	0 L 207	164	0 L 190	1 R 142
25	0 R 36	1 R 25	95	1 L 109	1 L 95	165	0 L 197	1 R 173
26	0 R 37	1 R 38	96	0 R 19	1 R 110	166	0 R 146	1 R 172
27	0 R 39	0 L 40	97	0 R 60	1 R 60	167	0 R 96	1 R 200
28	0 R 41	1 R 42	98	0 R 68	1 R 135	168	0 R 161	1 R 143
29	0 R 43	1 R 44	99	0 R 93	1 R 108	169	0 R 148	1 R 87
30	0 R 45	1 R 46	100	0 R 105	0 L 207	170	0 R 161	1 R 96
31	0 R 21	1 R 47	101	0 R 117	1 R 118	171	0 R 181	1 R 122
32	0 R 48	1 R 32	102	0 R 71	1 R 58	172	0 L 193	1 R 60
33	0 R 32	1 R 33	103	0 R 57	1 R 119	173	0 R 145	1 R 88
34	0 L 74	1 L 2	104	0 R 21	1 R 120	174	0 R 152	1 R 201
35	1 R 35	1 L 1	105	0 R 21	0 L 10	175	0 R 140	1 R 84
36	0 R 50	1 R 36	106	0 R 61	1 R 42	176	0 R 175	1 R 75
37	0 R 17	1 R 111	107	0 R 121	1 R 8	177	0 R 126	1 R 169
38	0 R 13	1 R 52	108	0 R 101	1 R 122	178	0 R 138	1 R 116
39	0 L 11	0 R 36	109	0 L 207	0 L 95	179	0 R 97	1 R 51
40	0 L 4	1 L 53	110	0 R 124	1 R 104	180	0 R 168	1 R 51
41	0 R 24	0 R 25	111	0 R 115	1 L 72	181	0 R 17	1 L 131
42	0 R 13	0 L 16	112	0 R 165	1 R 181	182	0 R 143	1 R 149
43	0 R 55	1 R 56	113	0 R 160	1 R 184	183	0 R 76	1 R 116
44	0 R 57	1 R 58	114	0 R 148	1 R 146	184	0 R 106	1 R 124
45	0 R 24	0 R 54	115	0 R 11	1 R 129	185	1 L 40	1 L 109
46	0 R 60	1 R 61	116	0 L 131	0 L 1	186	0 R 175	1 R 156

Continued on the next page

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
47	$0\,R\,62$	$0\,L\,16$	117	$0\,R\,39$	$1\,R\,72$	187	$0\,R\,182$	$1\,R\,183$
48	$0\,R\,25$	$1\,R\,48$	118	$0\,R\,61$	$1\,R\,132$	188	$0\,R\,66$	$1\,R\,204$
49	$0\,L\,35$	$0\,R\,63$	119	$0\,R\,61$	$1\,R\,133$	189	$0\,R\,201$	$1\,R\,197$
50	$0\,R\,16$	$1\,R\,50$	120	$0\,R\,70$	$0\,L\,16$	190	$0\,R\,84$	$1\,R\,191$
51	$0\,R\,147$	$1\,R\,143$	121	$0\,R\,14$	$1\,R\,132$	191	$0\,R\,157$	$1\,R\,186$
52	$0\,R\,67$	$1\,R\,68$	122	$0\,R\,21$	$1\,R\,134$	192	$0\,R\,64$	$1\,R\,7$
53	$0\,L\,40$	$1\,L\,53$	123	$0\,R\,139$	$1\,R\,189$	193	$0\,R\,150$	$1\,R\,127$
54	$0\,R\,22$	$1\,R\,54$	124	$0\,R\,137$	$1\,L\,20$	194	$0\,R\,18$	$1\,R\,94$
55	$0\,L\,11$	$0\,R\,50$	125	$0\,R\,130$	$1\,R\,180$	195	$0\,R\,104$	$1\,R\,96$
56	$0\,R\,60$	$0\,L\,16$	126	$0\,R\,203$	$1\,R\,156$	196	$0\,R\,90$	$1\,R\,88$
57	$0\,R\,69$	$1\,R\,70$	127	$0\,R\,75$	$1\,R\,176$	197	$0\,R\,156$	$1\,R\,199$
58	$0\,R\,61$	$1\,R\,71$	128	$0\,R\,75$	$1\,R\,205$	198	$0\,R\,140$	$1\,R\,123$
59	$0\,R\,54$	$1\,R\,59$	129	$0\,R\,202$	$1\,R\,157$	199	$0\,R\,150$	$1\,R\,112$
60	$0\,R\,49$	$0\,R\,25$	130	$0\,R\,12$	$1\,R\,141$	200	$0\,R\,151$	$1\,R\,198$
61	$0\,L\,28$	$1\,L\,72$	131	$0\,R\,30$	$1\,R\,69$	201	$0\,L\,200$	$1\,R\,168$
62	$0\,L\,117$	$1\,L\,30$	132	$0\,R\,39$	$1\,L\,124$	202	$0\,R\,196$	$1\,R\,28$
63	$0\,R\,73$	$1\,R\,63$	133	$0\,R\,153$	$0\,L\,16$	203	$0\,R\,163$	$1\,R\,193$
64	$0\,R\,97$	$1\,R\,150$	134	$0\,R\,154$	$0\,L\,16$	204	$0\,R\,146$	$1\,R\,177$
65	$1\,R\,185$	$1\,L\,53$	135	$0\,R\,88$	$1\,R\,136$	205	$0\,R\,161$	$1\,R\,194$
66	$0\,R\,144$	$1\,R\,145$	136	$0\,R\,196$	$1\,R\,140$	206	$0\,R\,45$	$1\,R\,100$
67	$0\,R\,76$	$1\,R\,208$	137	$0\,R\,24$	$0\,R\,48$	207	$0\,L\,4$	$0\,L\,207$
68	$0\,L\,196$	$1\,R\,31$	138	$0\,R\,99$	$1\,R\,37$			
69	$0\,R\,41$	$1\,R\,62$	139	$0\,R\,172$	$1\,R\,148$			

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